

Market Development and Efficiency in Emerging Stock Markets

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June 2001

Abstract

In 1990s emerging stock markets evolved from small and shallow into sizeable and liquid markets integrated with the world financial system. This paper empirically studies the conjecture that the relationship between market development and efficiency can be possibly captured by the weak-form market efficiency tests applied to moving subsample windows. The variance-ratio-based multiple comparison test (MCT) is applied to weekly and daily returns for 21 emerging stock markets over the 1988-2000 period. For each country, the MCT is used to test for random walk in subsample windows with fixed starting-point and moving end-points as well as those with fixed end-point and moving starting-points. Tests on both weekly and daily return series indicate that over time there is a move toward market efficiency. In those markets that showed rapid development, it becomes difficult to reject the random walk hypothesis as observations pertaining to earlier periods are dropped from the sample. However, in Mexico and the East Asian countries stock price behavior started to diverge from RW behavior soon after the reversal of portfolio equity flows during the financial crises that affected these countries.

JEL classification: G14, G15.

Key words: portfolio equity flows, random walk, variance ratio, multiple comparison test.

* Koç University, Istanbul, Turkey. email: kyilmaz@ku.edu.tr. I thank Emre Alper, Bülent Gültekin, Arie Melnick, Nejat Seyhun, Kemal Saatçioğlu, Barış Tan, and participants at the annual conference of the European Finance Association, the seminars at Bilkent University and Koç University. I would also like to thank Morgan Stanley Capital International for providing the data used in this paper.

I. Introduction

The objective of this paper is to empirically test whether the rapid development of emerging stock markets over the last decade helped these markets become more efficient over time. As the market size and liquidity increase, and the regulatory and the legal system evolve to operate more effectively, the systematic divergence of stock prices from the fundamentals should tend to diminish (see Demirgüç-Kunt and Levine 1996). Consequently, it becomes more difficult to predict future returns using publicly available information and markets move closer to the random walk behavior.

To be more precise, this paper asks the following question: Has the stock price behavior approached towards random walk as the emerging stock exchanges evolved through time from small, shallow and segmented markets into sizeable and liquid stock markets integrated with the international financial system? The answer to this question is crucial in developing a better understanding of the link between financial development and economic growth.

Since the mid-1990s there has been a growing body of literature analyzing various issues related to international portfolio equity flows, stock market development, volatility and comovement of emerging market stock returns. The significant degree of correlation between portfolio equity flows and the emerging stock market development indicators is established by the World Bank (1997). In a similar fashion, Demirgüç-Kunt and Levine (1996) show in a cross-sectional analysis that there is negative correlation between stock market development indicators and mispricing of stocks as measured by the deviation of prices from the price determined by an international version of the asset pricing theory.

In addition, several studies have shown that increased portfolio equity flows to emerging markets led to a decline in the cost of capital in these countries (Bekaert and Harvey 1995, 2000; Stulz 1997, Henry 2000a). Stulz (1997), on the other hand, shows that the stock return data did not support the hypotheses of increased stock return volatility and excessive comovement among emerging market stock returns. On the other hand, Subrahmanyam and Titman (1998) use their model of firm-financing decision to show that stock market growth leads to an increase in the information content of stock prices and this provides incentives for firms to go public. The decline in the cost of capital provides incentives for companies to raise long-term financing through the stock market, increasing

private investment expenditures and hence the rate of long-run economic growth (Levine and Zervos, 1998).

This paper can be viewed as part of a recent trend in the literature that investigates whether stock market development of the last decade has had any effect on market efficiency in developing countries (see Kim and Singal 2000, Kawakatsu and Morey 1999). However, these studies utilized exogenous or endogenous structural break tests, taking the exact date of stock market liberalization as the break point. This paper takes the view that financial liberalization is a necessary but not sufficient condition for market efficiency. Financial liberalization is the first step to attract international investors to a stock market. However, there may be other characteristics of the country (for example, political uncertainty or underdeveloped legal system) that will keep the foreign investors away. Only after foreign investors with large sums of investable funds get interested in the market, then the market may have a chance of becoming a sizeable and liquid market over time. Therefore, development indicators for a stock market will not necessarily start changing for the better immediately after the stock market liberalization. Furthermore, the move towards market efficiency need not take the form of a discrete jump in the underlying parameters of the stock price process. The move towards market efficiency can possibly be a continuous and a smooth process rather than a discrete change that takes place at a point in time.

In testing for market efficiency, the main tool of analysis is Lo and MacKinlay's (1988) heteroscedasticity-consistent variance ratio test, which is a test of the weakest form of the random walk hypothesis: Are stock returns serially uncorrelated at all leads and lags? Since the variance ratio test statistic is calculated for different return horizons, the multiple comparison test (MCT) is used to decide whether the VR test statistics for different holding periods together reject the random walk. The empirical analysis is based on daily and weekly Morgan Stanley Capital International (MSCI) stock price indices for 12 emerging markets that covers the period between January 1988 and March 2000 and daily MSCI indices for 9 other emerging markets for which the data have been available since January 1993.

As part of the empirical results the paper first shows that, for the majority of emerging markets analyzed in this study, the market development indicators have risen significantly in the first-half of the 1990s. The paper also documents the reversal in the

time trend of the market development indicators due to portfolio equity outflows in the aftermath of the recent financial crises.

Next comes the main focus of the paper. For each country, the MCT is applied to two types of moving subsample windows: with fixed starting-point, moving end-points; and with fixed end-point, moving starting-points. Both types of windows are used because they enable the researcher to identify the effect of each observation on behavior of the MCT statistic.

The rejection of the random walk hypothesis could be due to the underlying equilibrium model of stock returns, and therefore does not necessarily imply the rejection of the efficient market hypothesis. However, the paper tries to account for this possibility. After more than four decades of research on the subject, it is by now widely accepted that efficient market hypothesis should be viewed as a relative rather than an absolute concept (see Campbell, Lo and MacKinlay, 1997). Keeping this in mind, MCT results for the U.S. and Japanese markets are used as a benchmark with which to compare the results for emerging markets.

The paper is organized as follows. The next section presents the conceptual framework behind the analysis and, using annual market statistics for emerging markets, assesses the level of development in various markets over time. A brief account of the variance-ratio based MCT is provided in section III. Then section IV presents the empirical tests of random walk behavior in weekly and daily stock returns. Section V concludes the paper.

II. Market Development and Integration of Emerging Markets with the International Financial System

Until mid-1990s, almost all of the major papers written on stock price behavior and market efficiency have focused on industrial countries and especially on the United States. With the exception of several recent papers, little attention has been paid to stock price behavior in emerging markets.¹ Perhaps there was an implicit understanding that emerging stock market prices were bound to show significant divergence from random walk, as these markets lagged behind developed stock markets in terms of institutional infrastructure,

¹ Among these are Claessens, Dasgupta and Glen (1995), Harvey (1995), Korajczyk (1996), Bekaert and Harvey (1999, 2000), Kim and Singal (2000) and Henry (2000a, 2000b).

market size and liquidity. While there is some element of truth in this presumption, it ignores the diversity among emerging stock markets in terms of market development and institutional infrastructure. This viewpoint also overlooks the rapid pace of development in emerging stock markets since the late 1980's, which has been fueled by the surge in portfolio equity flows. According to International Finance Corporation's (IFC) Emerging Markets Factbook, market capitalization of emerging stock markets increased from \$110 billion in 1985 to \$2.14 trillion in 1996. During the same period, liquidity as measured by the total value of stocks traded in one year increased from \$43 billion to \$1.49 trillion. (For country level data see Table 6.1 of World Bank, 1997.)

While emerging markets in many respects differ from developed markets, still there is substantial diversity among emerging stock markets in terms of institutional infrastructure, market size and liquidity. For this particular reason, emerging stock markets provide a rich ground to see the link between market development and stock price behavior over time. This section presents and discusses development indicators for 12 emerging stock markets over the period from 1988 to 2000. The analysis follows the recent literature on emerging markets in constructing measures of stock market development².

Figure 1a presents the ratio of market capitalization over GDP, which is used as a measure of the market size. Two measures of market liquidity are presented in Panel (b) and (c) of Figure 1. The first is the ratio of the value of all stocks traded throughout the year over GDP. This indicator is intended to measure the liquidity of a market in relation to the size of the economy. Turnover ratio is the second measure of liquidity. It measures liquidity relative to the value of all shares outstanding and is inversely related to transaction costs in the stock market. Both measures of liquidity are included in the analysis, because, as it is clearly shown in the new database prepared by Beck *et al* (2000), the two measures do not necessarily move in the same direction. Finally, as a measure of the prevalence of public equity financing among domestic companies, Figure 1d presents the number of domestic companies whose stocks are traded in the stock market. Three-year averages of these indicators are presented in Table 1.

There are several important observations to make about the behavior of market development indicators over time. First, it is evident in both Figure 1 and Table 1 that the

majority of emerging stock markets went through a rapid transformation since 1988. In almost all countries, stock market size and liquidity has increased in the late 1980s and the early 1990s. Market size and liquidity indicators peaked between 1993-95. In a matter of less than a decade, many emerging markets have attained market size and liquidity measures that were close to many mature markets.

Second, there is a discernible heterogeneity among emerging markets in terms of market development. For example, according to all four measures of market development, Taiwan, Korea and Malaysia could easily be classified among mature markets as early as 1988. It is true that these markets have been positively affected by portfolio equity flows and even after the East Asian crisis both market size and liquidity is comparable with mature markets.

A second line of division runs through regions. The increases in the size, liquidity and number of companies are more pronounced in the Asian countries than in Latin American countries. Some East Asian countries, such as Thailand, Indonesia and Philippines, were rather under-developed in the late eighties with market capitalization/GDP and traded value/GDP ratios close to or below 10%. In the late 1980's and early 1990's, these countries' stock markets developed significantly, mostly thanks to portfolio equity flows. A similar stock market development performance, albeit at a smaller scale, could be observed in the majority of the emerging market economies in the sample, such as Argentina, Brazil, Chile, Mexico, Greece and Turkey. These countries went through a steep upward trend in market capitalization/GDP and traded value/GDP ratio, to be reversed in or after 1995.

There are several factors to explain the unprecedented development of emerging markets. The first is related to economic reforms in developing countries. Many countries have undertaken financial reforms and liberalization during the late 1980s. Financial reforms are likely to increase the flow of domestic savings to stock market.

Another and perhaps a more important factor that contributed to the rapid development of emerging stock markets was the rapid increase in portfolio equity flows

² See Beck, Demirguc-Kunt and Levine, 1999 and World Bank, 1997.

from industrial to developing countries.³ While in 1988 only \$1.1 billion portfolio equity capital flowed into emerging equity markets from industrial countries, this amount reached \$51 billion in 1993 and stayed close to \$40 billion until the Asian crisis of 1997. Since then portfolio equity flows slowed down to as low as \$15.6 billion in 1998 along with commercial loans and international bond issues.

As highlighted by Stulz (1997) and empirically shown by Bekaert and Harvey (1995) and Henry (2000a), the financial capital flows from industrial to developing countries are to the benefit of both sides. Investors from industrial countries are expected to gain as they earn a higher return to their capital in emerging markets. Emerging market economies, on the other hand, are also expected to gain, because the availability of these funds will reduce the cost of capital, enhance investment in longer-term projects and increase the long-run growth potential of the country. Bekaert and Harvey (1995) showed that with the integration of a market to world financial system, the cost of capital is no longer related to the variance of the local stock index return only. It is related to the covariance of local stock market return with world market returns, which is expected to be lower than the variance of the local stock returns.

In addition to direct effects, increased portfolio equity flows to emerging markets may have indirect effects on the size and liquidity of emerging stock markets as well. These effects can be instrumental through several channels. First, as foreign investors become actively involved in the domestic stock market, stock prices would increase, encouraging domestic firms to raise funds through initial public offerings in the stock market. Consequently, the size of the stock market will increase both in terms of total market value of stocks and in terms of the number of listed companies in the stock market.

The activities of the foreign investors may also increase the liquidity of the market. As there are more investors with sizeable investment funds in the market, it becomes easier to find buyers or sellers in the market. Furthermore, to the extent that there are economies of scale in financial intermediation, increased market liquidity leads to a decline in transaction costs attracting more domestic investors to the market. As the size and liquidity of the market increase over time, the ability of a few large domestic investors to manipulate stock prices would diminish. This, in turn, will lead to an increase in public

³ According to the estimations reported in the World Bank (1997), the correlation coefficients between the portfolio equity inflows/GDP ratio and the market capitalization/GDP ratio and between the portfolio equity

confidence in the stock market, making the stock market a more attractive investment alternative for individual investors.

Another channel through which the portfolio equity flows may help increase the public confidence in the stock market is the regulatory reform. Once the emerging country governments realized the opportunities generated through foreign capital flows, they tried their best to attract more foreign financial capital flows. However, one important criterion for foreign investors in choosing the destination country is the soundness and effectiveness of the regulatory system and the quality of financial intermediation. Realizing this relationship, many governments actively pursued policies to improve the capital market regulatory framework and elevating domestic accounting and supervision standards to international levels. The regulatory reform definitely helps increase public confidence in the market and attract more domestic investors to it. The result would be an increase in the market size and liquidity, providing more investable funds to finance longer-term projects (see World Bank, 1997).

Finally, the positive impact of increased portfolio equity flows on emerging markets will work its way out through the increase in the information content of stock prices. The portfolio equity flows will enhance both the quantity and the quality of information available about emerging stock markets and individual stocks. The majority of foreign investors are institutional investors with large portfolios. Losses resulting from inadequate information can be quite high for these investors. Furthermore, the cost of purchasing information is quite low relative to the sums they invest in a market. Consequently, with the flow of portfolio equity flows to an emerging market, the demand for financial information in this market is expected to increase. In response, brokerage firms devote more resources to information gathering and processing. This will increase information about the individual stocks, aggregate economy and the sector not only to foreign investors, but domestic investors as well. Due to economies of scale in the information processing, market information will become more affordable for institutional and individual domestic investors, moving the market closer to efficiency (see World Bank, 1997, chapter 3).

Last three years have shown that portfolio equity flows were reversible. There is evidence that the reversal of portfolio equity outflows increased the volatility of the

flows/GDP ratio and the trading volume/GDP ratio are both 0.71.

emerging market stock returns. Portfolio equity outflows also raised the question of what impact it will have on stock market development. Will markets lose all the gains in terms of size, and liquidity or are some of the favorable effects of foreign equity investment on emerging stock markets irreversible?

With the portfolio equity outflows there was a decline in market size and liquidity in emerging stock markets. However, the number of listed companies did not decline significantly. Even so, the decline in the market development indicators is expected to be temporary. This is so, because the reversal in portfolio equity flows does not necessarily mean that the integration of the emerging markets to the international financial system will be reversed. Unless the host country governments take measures to limit the free flow of financial capital as did the Malaysian government, the integration process will continue. As the tremors of the financial crises subsided, international investors started to move their money back to emerging stock markets. As it's already been mentioned above, portfolio equity flows to emerging markets increased from \$16 billion in 1998 to \$28 billion in 1999.

It would not be wrong to expect that the new round of portfolio equity flows to emerging markets will be more robust for two reasons. Over time and especially after the financial crises in emerging markets, international investors learned more about the individual markets and stocks. Therefore, they will be more selective about the markets or individual stocks. Secondly, in the post-crisis period many countries started to address the weaknesses in their financial system more carefully and put more effort to reform them. These efforts in emerging markets will help create an environment that is more conducive to foreign investor participation.

III. Empirical Analysis: Multiple Comparison Test on Moving Windows

Weak-form market efficiency hypothesis is tested through standard test procedures. In these tests, it is common to assume that the expected return is known and constant. With this assumption in place, the test for random walk behavior is identical with the test for weak-form market efficiency. There are several versions of the random walk hypothesis. Following the recent empirical literature on market efficiency, this paper focuses on the weakest form of the random walk hypothesis. Namely, it tests for uncorrelated (but not independent) increments of stock prices.

All tests of the weak-form random walk hypothesis are related to the correlogram of stock returns and evaluate if the autocorrelation coefficients at different lags and leads are equal to zero. Since tests of random walk do not run the risk of misspecification of the alternative hypothesis, they are expected to be theoretically reliable. In practice, however, they were found to have quite low statistical power. Consequently, even when stock prices follow random walk behavior, random walk hypothesis can be on occasion rejected. Among the random walk tests that rely on alternative representations of the autocorrelogram of the return series are variance ratio test (Cochrane 1987, Lo and MacKinlay 1988, and Poterba and Summers, 1988), Fama and French's (1988) regression test and Jegadeesh's (1991) regression test.⁴

Variance ratio test is a weighted-sum of estimated autocorrelation coefficients, with the weights declining in return horizon. Being a two-tailed test, the rejection by the variance ratio test reveals information about autocorrelation structure. If the variance ratios at different return horizons are greater (smaller) than one, then one cannot reject the hypothesis that stock returns are positively (negatively) serially correlated. Accordingly, a value of one for the variance ratio means that stock prices follow random walk. Random walk series possesses a unit root and random walk increments are required to be uncorrelated.

Lo and MacKinlay's heteroscedasticity-consistent variance ratio test statistic, $z_2(q)$, is used to test the null hypotheses for a specific aggregation value (holding period), q . If $z_2(q)$ is greater than the critical value of the standard normal distribution, then the random walk hypothesis is rejected. However, because variance ratios may be statistically close to 1 for some q , and different from 1 for others, variance ratio test procedure based on various holding periods can easily lead to inconclusive results.⁵

Chow and Denning (1993) proposed a solution to this problem. They used Hochberg's (1974) theorem on multiple comparison tests and extended Lo and MacKinlay variance ratio test to incorporate joint comparison of selected variance ratios with one.

⁴ What differentiates Lo and MacKinlay's variance ratio test from others (mainly from the one used by Cochrane, 1987, and Poterba and Summers, 1988) is that it is possible to conduct the test even when stock returns are heteroscedastic.

⁵ Lo and MacKinlay (1989) reported Monte Carlo calculations showing that variance ratio test had better size and power performance compared to two alternative unit root test, namely, Dickey-Fuller's t -test and Box-Pierce portmanteau Q -test. In other words, when the focus is the absence of correlation among the increments the variance ratio test is preferred.

Taking k Lo-MacKinlay variance ratio test statistic, $z_2(q_i)$ s, for $i=1,2,\dots,k$, that have standard normal distribution under the random walk null hypothesis, and using Bonferroni probability inequality, Hochberg (1974) test amounts to picking up the test statistic $z_2(q_i)$ with the largest absolute value. Hochberg (1974) showed that the largest absolute value of these k standard normal variates has a Studentized Maximum Modulus (SMM) distribution with k and nq (sample size) degrees of freedom.

Critical values for the SMM distribution are greater than the critical values for the standard normal distribution. This is not surprising given that the MCT statistic is the maximum of the variance ratio test statistic for all selected holding periods. Chow and Denning (1993) applied MCT to U.S. stock market indices between 1962-1985 and found statistical evidence to reject the random walk hypothesis for the equal-weighted index, but not for the value-weighted index.

Previous studies of random walk behavior conducted tests for a given fixed sample period. This sample period is either dictated by the availability of data or by visible breaks such as the Great Depression, the Second World War, the policy switch from fixed to flexible exchange rate regimes in major industrial countries or, for that matter, the financial liberalization in emerging markets. So far, no study has focused on the possibility of a smooth change in the behavior of stock prices over time. This is not surprising. Most of the research on market efficiency and random walk behavior has focused on already developed stock markets and market development through time has never become an issue.⁶

The same approach cannot be justified in the case of emerging markets. As it's already shown above, these markets have gone through a period of rapid development largely due to the unprecedented increase in international portfolio flows. If stock market development resulting from increased international capital flows is expected to have a significant impact on stock price behavior, it may not be possible to capture this effect by a test of random walk behavior using subsamples starting in mid- or late eighties. A sample period of 1988-2000 would not be suitable to analyze the move towards market efficiency. If the increase in international capital flows had any impact on stock market development,

⁶ Even in the case of mature stock markets, the possibility of a change in the behavior of stock returns over time sparked by unobservable factors cannot be ruled out. Yet, there has been no study testing for random walk or some other hypothesis with moving subsample windows.

it would not be surprising to observe stock price series deviating from random walk behavior in the 1988-92 sample and moving towards random walk behavior in the 1993-2000 subsample. Applying MCT to this series over the full 1988-2000 sample may lead to the opposing effects of the two subsamples, the final outcome of which is not possible to discern theoretically. If the regime switch were known a priori, the correct approach would be to apply MCT to two subsamples separately. Since this is not the case and the available techniques may not help one in detecting the regime switch accurately, a reasonable approach is to apply MCT to moving windows of data series.

In a recent paper, Kawakatsu and Morey (1999) study the impact of financial liberalization on market efficiency in 16 stock markets. Their tests focus on the market efficiency before and after the liberalization. Using structural break tests, they are not able to obtain evidence suggesting that financial liberalization led to improvement in the efficiency of emerging markets. Kim and Singal (2000) also analyze the stock price behavior before and after the stock market liberalization in 14 emerging markets. Their work is close to this study in terms of methodology, in that they use Lo and MacKinlay's variance ratio test. Their results are not in line with the findings of Kawakatsu and Morey (1999). Using the variance ratio test, in the majority of the countries considered they showed that the variance ratio drops close to one when they move from the pre-liberalization period to post-liberalization period.

Completely different results obtained by Kawakatsu and Morey (1999) and Kim and Singal (2000) may follow from the choice of different liberalization dates. For many countries their choice of liberalization dates do not coincide. Table 2 provides a list of liberalization dates for each country published in four different sources (Bekaert and Harvey 2000, Buckberg 1995, Kim and Singal 2000, and IFC 1990-97) and compiled by Bekaert and Harvey (2000). As can be clearly seen from the table, there are many countries where four different sources are in disagreement for the exact date of liberalization.

Besides, as it's argued above, stock market liberalization is a necessary but not sufficient condition for the move towards market efficiency. Therefore, a structural break does not necessarily take place immediately after the financial liberalization. Furthermore, there does not need to be a discrete jump in the underlying parameters of the model for a

market to become efficient. Therefore, one should not disregard the possibility that the move towards market efficiency is a smooth and continuous process.

In the empirical analysis two types of moving windows are considered. In the first type of window, the starting-point of the sample is fixed in the first week of January 1988 and MCT statistic for the first 250 weekly (750 daily) observations is calculated. Then, moving the end-point of the subsample one-week (one-day) forward while holding the starting-point fixed the sample period is expanded by one observation at a time and the MCT statistic is calculated again. This exercise was repeated until the full sample size is reached and the MCT statistic for all subsamples considered are plotted (MCT_i , $i = 1, 2, 3, \dots, T-249$). Adding new observations to the subsample is not expected to lead to a significant change in MCT results as long as the observations pertaining to low development phases are kept in the sample.

Another way of forming a sequence of moving windows is to hold the end-point fixed in the last week of March 2000 and move the starting-point forward starting from the first week of January 1988 until the sample size declines to a minimum of 250 weekly (750 daily) observations. Again, the MCT statistic for all subsamples considered (MCT^i , $i = 1, 2, 3, \dots, T-249$) are plotted.⁷

We apply our empirical methodology to a total of 21 emerging stock markets. The data for 12 markets span the period from January 1988 to March 2000, for 9 emerging markets from January 1993 to March 2000. For all emerging stock markets the stock market indices from the Morgan Stanley Capital International (MSCI) database are used. The data set for the U.S. stock markets includes, Dow Jones Industrials Average and the value weighted index of all NYSE and AMEX stocks from CRSP tapes (see the data appendix).

In the empirical analysis weekly and daily rather than monthly stock returns data are used. Monthly stock price series are not used because that would reduce the number of observations significantly, the use of which will lead to small sample bias in the multiple comparison test. Following Lo and MacKinlay (1988), random walk is tested in weekly Wednesday-to-Wednesday real stock returns. Being closest to a definition of a week,

⁷ A third alternative is to test for random walk for moving fixed-length subsample windows. The results for this test are not reported, because it cannot differentiate the impact on the multiple comparison test statistic

weekly Friday-to-Friday closing real stock returns are also used, even though they are likely to be influenced by the weekend effect.

Before the MCT results for the moving subsample windows with fixed end-points and fixed starting-points, first descriptive statistics for real weekly Wednesday returns are presented in Table 3 to highlight the characteristics of stock return series for the full sample. Average weekly real stock returns vary from -0.03 percent in Thailand Korea to 0.39 percent in Argentina. In absolute terms they are not much different from NYSE and Tokyo Stock Exchange in this regard. The difference between emerging and mature markets is more apparent in terms of the measures of volatility. Standard deviation of real stock returns in emerging markets are two-to-four times the standard deviations of returns in the New York and Tokyo stock exchanges. Emerging market real stock returns are skewed and display excess kurtosis. Jarque-Bera test rejects normality for all markets.

Table 4 presents the variance ratios and heteroscedasticity-consistent variance ratio test statistic (z_2) for 2, 4, 8, 16 and 32 weeks holding-periods as well as the MCT statistic for the full sample. To give an example, the MCT statistic for Indonesia is 1.858 whose absolute value is the highest among the five VR test statistic calculated. However, MCT does not reject random walk for Indonesia. MCT rejects random walk behavior for Philippines, Thailand, Chile, Greece and Turkey at the 5 percent and for Taiwan at the 10 percent significance level. With this result, one could conclude that the MSCI indices for these emerging stock markets do not follow random walk. In contrast, the MCT statistics for Indonesia, South Korea, Malaysia, Argentina, Brazil and Mexico do not reject the random walk for these markets. Furthermore, daily returns for DJIA and Nikkei 225 indices show no deviation from random walk during the 1988-1999 period.

It is too soon to reach a firm conclusion about the behavior of stock prices in emerging markets that are included in the sample. As already mentioned above, it is possible that the rejection of random walk in the full sample is driven by the behavior of stock prices in the earlier parts of the sample period. On the other hand, the inability to reject random walk for the full sample does *not* necessarily imply that the behavior of stock returns throughout the 1988-2000 period can be characterized by random walk behavior.

of the observation dropped from the sample and the observation added to the sample. In addition, test results will be dependent on the choice of the length of the window.

Plots of the MCT statistic for subsample windows with fixed starting-point and those with fixed end-point will be helpful in deciding whether there has been any move towards random walk behavior. If there has been a move towards random walk behavior throughout the January 1988-March 2000 period, this would be captured by MCT applied to the subsample windows. The MCT statistic is expected to decline as the starting-point moved from January 1988 forward until a minimum sample size of 250 weekly observations is reached.⁸ Prices in stock markets with low levels of development in the late 1980s are less likely to follow random walk. Consequently, the MCT statistic for the subsample windows including earlier parts of the sample will be high, possibly rejecting the random walk. If these markets had experienced a move towards market efficiency throughout the 1988-2000 period, MCT statistic will decline as the earlier periods are dropped from the subsample and at some point will fall below the critical value.

In the case of subsample windows with fixed starting-points, no drastic change in the MCT statistic is expected as the end-point of the subsample window is moved forward, including more recent observations in the sample.⁹ As long as earlier periods are retained in the sample, test results are not likely to capture the move towards random walk behavior. As more recent observations are added to the sample, the resulting time series variable will be a mixture of both “behavioral regimes” in stock prices. Consequently, it will be quite difficult to detect the change in the price behavior using the variance ratio or other tests of random walk.

Before going into the moving-window analysis of the MCT statistic for emerging market returns, Figure 2 plots the MCT statistic for the daily nominal returns for the Dow Jones Industrials Average (DJIA) and the Nikkei 225 indices. Figure 2 plots both the subsample windows with fixed starting-point and fixed end-point for the two indices, respectively. Each dark (light) dot represents the MCT statistic for moving windows with fixed starting-point (end-point). For DJIA, MCT statistic for both types of windows are below the 5% critical value (upper bold straight line) tabulated in Hahn and Hendrickson (1971). In the case of Nikkei 225, however, there is a slight increase in the MCT statistic towards the 10% critical value (lower straight line) as the starting-point of the fixed end-point window is moved towards the mid-1995 and beyond. This is the period of the worst

⁸ Depending on the holidays, the starting-point of the minimum sample with 250 observations is somewhere in May or June 1995.

economic and financial crisis in the Japanese history. Yet, the rejection of the random walk behavior for daily returns is not too strong. Even with the daily returns it is difficult to reject the random walk hypotheses for the DJIA and Nikkei 225 indices. When weekly returns are used, neither of the two stock markets shows any deviation from the random walk, indicating empirical support for random walk behavior in the New York and Tokyo stock markets.

The coverage of DJIA is limited to 30 blue chip stocks. For that reason, the MCT procedure is applied to daily and weekly Wednesday returns of the value-weighted CRSP NYSE-AMEX index. Lo and MacKinlay (1988) applied their variance ratio test statistic to weekly observations for the 1962-1985 period, and rejected the random walk. As can be seen in Figure 3, the MCT statistic rejects the random walk hypothesis for the starting-point fixed windows until the observations for the post-1987 crisis (October 17, 1987) are included in the data set. Test results fail to reject the random walk hypothesis for all subsample windows with the end-point fixed in May 1999 and the starting-point is moved from July 1962 onwards. This result shows that Lo and MacKinlay's rejection of the random walk hypothesis is mostly driven by the observations pertaining to the earlier part of the sample.

Next is the analysis of emerging market stock returns. MCT statistics for fixed starting- and fixed end-point windows and weekly Wednesday and Friday real stock returns of 12 emerging markets are plotted in Figures 4 and 5, respectively. There is no uniformly conclusive evidence that would verify the initial conjecture about the behavior of the MCT statistic for both types of windows.

Both Wednesday and Friday return results for S. Korea, Argentina, Brazil and to some extent Mexico appear to support the random walk behavior in these markets throughout the full sample. MCT statistic for both types of windows lie beneath the 5 and 10 % critical values for all subsamples considered (Figures 4 and 5). Among these markets, however, only South Korea had high levels of market development indicators throughout the 1988-2000 period. Argentina, Brazil and Mexico's stock markets had shown rapid degrees of development during this period.

⁹ The end-point of the minimum sample with 250 observations is somewhere in October or November 1992.

The evidence for Philippines, Chile, Greece and Turkey supports the initial conjecture that emerging stock markets become more efficient over time. MCT statistics for these countries are always above the critical value in the case of subsample windows with fixed starting-point. However, the MCT statistics for the fixed end-point windows tend to decline and, eventually, MCT fails to reject random walk behavior for these countries as earlier observations of stock returns are dropped from the sample.

The effect of the recent crises on some countries can be observed in the plotted MCT statistic. With the exception of Taiwan, which was the least-affected Asian country during the East Asian crisis, the MCT statistic for Wednesday as well as Friday returns shut up immediately after the observations pertaining to that period are included in the fixed starting-point window. In the case of Malaysia (Friday) and Thailand (Wednesday) the jump was sufficiently big to move the test statistic to the critical region. The effect of the Mexican crisis on Mexican stock returns is also evident from the plots of MCT statistic for the fixed starting-point window (see Figures 4 and 5).

Test results for the weekly Wednesday returns of Indonesia stand out. There is a big jump in the test statistic for the fixed end-point windows as the observations are dropped from the beginning of the subsample one at a time. A similar jump is not observed in the case of fixed starting-point windows. One can interpret this result as indicating that Indonesian stock market used to be efficient in the past, but not in the 1990s. However, it is too early to reach at this conclusion. First, a similar pattern is observed for Wednesday returns only and not for Friday returns. Second, the big jump in the MCT plot may be due to some outlier stock return observations. For example, while the real weekly Wednesday return was 80% in the week of December 21 1988, just one week later it went down to -36%. It is quite possible that putting these two extreme values one after the other with the rest of the data set can possibly change the test results. While the rest of the returns tend to have a positive correlation among each other, these two observations work fully against it, because they are negatively correlated.

One way to understand whether the outliers have such a significant impact on test results is to apply the test after dropping the outliers. In deciding what is an outlier, the

absolute value of the cutoff point is set at 20%, above which less than a handful of returns are available. Test results after dropping the outliers are presented in Figures 6 and 7.¹⁰

As suspected, there are significant changes in the test results once the outliers are dropped. First, qualitative differences among the test results for Wednesday and Friday returns diminished substantially. Now MCT rejects the random walk for Indonesia irrespective of the day of the return or the type of the subsample window. Second, the results for Philippines, Taiwan, Thailand, Chile, Mexico, Greece and Turkey are consistent with the initial conjecture. As the earlier observations are dropped from the end-point fixed subsamples for each of these countries, the tendency to reject the random walk diminishes. On the other hand, with the exception of Mexico, the random walk is rejected for these countries as long as the earlier observations are included in the fixed-starting point sample.

Third, test results for Mexico and the crisis stricken countries of East Asia (except for Taiwan) identify the tendency of these emerging market stock indices to deviate from random walk as they started to experience substantial portfolio outflows during crisis. The plotted MCT statistic for the fixed starting-point subsample window displays a significant upward move as the observations pertaining to the crisis period are included in the data set. In the case of Malaysia, which imposed restrictions on international capital flows in the October 1998, this increase is sufficient to reject the random walk. For the same reason, MCT rejects the random walk for Malaysia as long as the subsample windows include the data after the East Asian financial increases.

The analysis of emerging market stock price behavior is not restricted to weekly returns only. After analyzing the results for weekly return series, it is crucial to emphasize one observation about the plots of MCT statistic for weekly returns. Even after dropping the outliers from the data, there is substantial fluctuation in the plots of the MCT statistic for both fixed starting- and end-point windows. One reason for this fluctuation could be the fact that Wednesday and Friday data are not always available because of holidays. If a stock market is closed on a particular Wednesday (Friday), then as a rule for that week Tuesday or Thursday's (only Thursday's) index level is used. Consequently, the weekly return is calculated on a short or long week, which can possibly create extra fluctuations in the MCT statistic. These effects may not be observed in the case of daily data. For this

¹⁰ Outlier return observations as well as their dates are presented in Table A1.

reason, it is useful to apply the MCT to daily stock market returns.¹¹ There is a second reason for using daily returns. Using daily data increases the number of observations available and therefore allows one to exploit the large sample properties of the variance ratio test not only for those countries with observations starting in 1988, but also for other emerging markets whose MSCI stock index data started in 1993.

The problems associated with the use of daily returns are well known. Using daily returns for the S&P 500 index, Lo and MacKinlay (1988) show that one is likely to capture spurious correlation caused by infrequent trading of especially small and illiquid stocks. Lo and MacKinlay's conclusion about the spurious correlation in daily returns is supported by the MCT statistics for nominal daily returns of the value-weighted CRSP NYSE-AMEX index plotted in the right panel of Figure 3. However, even with the presence of spurious correlation, the MCT statistic for the fixed end-point windows goes down and eventually falls below the critical region in the 1990s as earlier observations are dropped from the subsample with fixed end-point.

Despite the evidence by Lo and MacKinlay (1988) on spurious correlation caused by small stocks, random walk hypothesis for emerging stock markets is also tested using daily MSCI indices. MSCI indices include only large and liquid stocks about which substantial information available to international investors. Therefore, there is likely to be little spurious correlation caused by infrequent trading of some stocks.

Figures 8 and 9 plot the MCT statistics for subsample windows with fixed starting- and end-point for the 1988-2000 and 1993-2000 samples, respectively. In calculating the MCT statistic 2, 4, 8 and 16-day holding periods are used only.

The results for the daily returns are in general consistent with the results for the weekly returns. In the case of subsample windows with fixed end-points, the MCT statistic for the majority of emerging markets display a downward trend and eventually fall below the critical value. The MCT statistic for all countries, except S. Korea, (Figures 8 and 9) display a downward trend as observations are dropped from the beginning of the sample one at a time with the end-point fixed. Among these, for Taiwan, Argentina, Brazil, Mexico, Turkey (Figure 8) and India, Pakistan, Venezuela and South Africa

¹¹ The sensitivity of the test results for daily returns on the use of nominal rather than real or excess returns is checked for Turkey. Turkey has been one of the few chronic and high inflation countries. Daily inflation

(Figure 9), the downward trend continues until the test statistic falls below the critical value. Although the MCT statistics for Indonesia, Philippines, Thailand, Chile, Greece (Figure 8) and China, Sri Lanka, Colombia, and Peru (in Figure 9) fall as the starting-point is moved forward, they do not fall below the 5 % critical level. MCT statistics for the fixed end-point windows of Malaysian and Israeli stock returns always stay below the critical level in all subsample windows considered.

In the case of windows with fixed starting-point, the MCT statistic stay above the critical value for all subsamples and countries considered, except for S. Korea and Israel. In the case of Malaysia and Thailand, the test statistic is below the critical value to start with, but moves above the critical region as the observations pertaining to the post-Asian crisis are included in the subsample.

In the case of subsample windows with fixed starting-point, a reversal in the trajectory of the MCT statistics is observed in Figures 8 and 9. MCT statistics tend to increase as more recent observations are added to the end of the sample. However, in the majority of the countries the MCT statistic is always above the critical level irrespective of the subsample used. The exceptions are Israel and S. Korea. Israel and S. Korea differ from other countries in that the MCT was unable to reject the random walk behavior for all subsamples with fixed starting-points considered. In the case Malaysia and Thailand the MCT statistic tend to be slightly lower than the critical level when data pertaining to the pre-crisis period are used. However, the MCT statistic increases above the critical level to reject the random walk behavior as new observations are added to the end of the sample. Overall the evidence from daily nominal returns is in support of the results of the weekly real return series.

IV. Conclusions

This paper investigates the link between market development and efficiency in emerging stock markets since 1988. First, it shows that for the majority of emerging markets analyzed, the market development indicators have risen significantly between mid-1980s and mid-1990s, to be partially reversed in the second half of 90s. The rapid pace of

rates are obtained from the monthly inflation figures. Since the recorded inflation rate varies much less frequently than stock returns, there is no spurious correlation imposed by ignoring the inflation rate.

development in the early 1990s and the ensuing reversal in emerging stock markets were partly due to the changes in international capital flows to these markets.

After presenting available data on market development, the paper uses variance-ratio-based MCT to test for random walk behavior in emerging stock market indices over time. Random walk behavior is tested using weekly and daily stock returns for 21 emerging markets between January 1988 and March 2000. The empirical methodology is to test for random walk behavior by applying the MCT to two types of subsample windows: One with fixed end-points, and one with fixed starting-points. Test results for the subsample windows with fixed-end points show that in some of the emerging markets that developed rapidly over the last decade, it becomes rather difficult to reject the random walk hypothesis as the observations pertaining to earlier periods are dropped from the subsample.

The plots of the MCT statistic for windows with fixed starting-points stand in contrast to those for windows with fixed end-points. There is no discernible downward trend in the MCT statistic as more observations are added to the end of the sample at a time. In the majority of the emerging stock markets, the MCT statistic rejects the random walk for all windows with fixed starting-points.

Combined with the results on market development, plots of the MCT statistic indicate that emerging market stock indices converge towards random walk behavior as these markets evolve through time from small, shallow and segmented markets into sizeable and liquid stock markets integrated to the world financial system. In the case of American and Japanese markets and those emerging markets with characteristics similar to mature markets in the late 1980s, stock prices appear to follow random walk irrespective of the subsample window used.

The results also show that financial crises such as the Mexican and the East Asian crises adversely affected the emerging markets' ability to price stocks efficiently. Finally, the results for Malaysia clearly indicate that as a result of the imposition of capital controls stock price behavior might diverge from random walk.

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Table 1. Market Development Indicators

Values in columns 2 thru 5 and 8 thru 11 are the average values of the corresponding indicator over the period indicated in the column 6 for each country. The number in the second column is the average of the number of firms whose stocks are traded in the stock market. Column 3 and 4 (and columns 9 and 10) give the average of the ratio of the stock market capitalization and the total value of all stocks traded throughout the year over GDP. Finally, the numbers in column 5 (and column 11) are the average of the ratio of column 4 over column 3 and is a measure of liquidity in the market.

	No of Firms	Market Cap/GDP	Traded Value/GDP	Turnover Ratio	Average of		No of Firms	Market Cap/GDP	Traded Value/GDP	Turnover Ratio
Indonesia	24	0.001	0.00003	0.027	1985-87	Argentina	217	0.022	0.005	0.223
	108	0.049	0.021	0.387	1989-91		177	0.059	0.019	0.325
	182	0.271	0.066	0.248	1993-95		170	0.128	0.032	0.222
	283	0.275	0.145	0.738	1997-99		130	0.211	0.083	0.455
S. Korea	362	0.151	0.110	0.701	1985-87	Brazil	1215	0.135	0.079	0.586
	660	0.466	0.379	0.811	1989-91		1193	0.081	0.028	0.344
	693	0.424	0.598	1.343	1993-95		1136	0.229	0.148	0.564
	737	0.405	0.882	2.574	1997-99		1060	0.273	0.202	0.773
Malaysia	226	0.550	0.079	0.143	1985-87	Chile	223	0.209	0.015	0.065
	285	1.145	0.221	0.193	1989-91		216	0.534	0.037	0.072
	419	2.651	1.711	0.559	1993-95		262	0.993	0.109	0.097
	747	1.420	0.806	0.707	1997-99		286	0.863	0.084	0.096
Philippines	135	0.059	0.023	0.321	1985-87	Mexico	167	0.042	0.051	1.041
	153	0.213	0.039	0.185	1989-91		204	0.193	0.063	0.324
	180	0.622	0.180	0.223	1993-95		197	0.436	0.176	0.442
	224	0.489	0.223	0.474	1997-99		191	0.312	0.095	0.306
Taiwan	133	0.280	0.381	1.144	1985-87	Greece	115	0.049	0.004	0.050
	200	0.980	4.361	4.700	1989-91		130	0.177	0.035	0.177
	285	0.805	2.061	2.233	1993-95		163	0.161	0.057	0.307
	450	1.087	3.873	3.474	1997-99		263	0.866	0.706	0.724
Thailand	108	0.076	0.045	0.515	1985-87	Turkey	82	0.037	0.0013	0.036
	222	0.333	0.253	0.774	1989-91		107	0.098	0.034	0.323
	347	0.829	0.532	0.560	1993-95		158	0.145	0.202	1.366
	405	0.312	0.216	0.761	1997-99		281	0.363	0.356	1.224

Source: IFC Emerging Market Database

Table 2. **Financial Liberalization dates**

Country	Bekaert-Harvey (2000)*	Buckberg (1995)	IFC EM Factbook	Kim-Singal (2000)
Indonesia	Sep. 90/Sep. 89	Mar. 89	Sep. 89	Sep. 89
S. Korea	Sep. 88/Jan. 92	--	Jan. 92	Jan. 92
Malaysia	Dec. 88	--	Dec. 88	--
Thailand	Dec. 88/Sep. 87	--	Dec. 88	Aug. 88
Taiwan	Jan. 91	--	Jan. 91	Jan. 91
Philippines	Jun. 91/Nov. 91	Oct. 89	Oct. 89	Mar. 86
Argentina	Dec. / Nov. 89	Oct. 91	Oct. 91	Nov. 89
Brazil	May 91	May 91	May 91	May 91
Chile	Apr. 90/Jan. 92	Oct. 89	Dec. 88	Oct. 89
Mexico	May 89	May 89	May 89	May 89
Greece	Dec. 87	--	Dec. 88	Aug. 86
Turkey	Aug. 89	Dec. 89	Aug. 89	Aug. 89
Colombia	Feb. 91	Oct. 91	Feb. 91	Feb. 91
Peru	--	--	--	--
Venezuela	Jan. 90	Dec. 88	Jan. 90	Jan. 90
India	Nov. 92	--	Nov. 92	Nov. 92
Pakistan	Feb. 91	June 91	Feb. 91	Feb. 91

* The first date is given in the NBER working paper version (1997), whereas the second date is the one listed in the final version of the paper

Table 3. **Descriptive Statistics for weekly Wednesday real stock returns**

Columns 2 through 7 present the country acronyms used throughout the paper, the sample mean, median, maximum, minimum and standard deviation of the stock returns. The numbers in columns 8 and 9 are the skewness and kurtosis statistics for each stock market returns. Columns 10 and 11 give the Jarque-Bera test statistic for the normality of stock returns and the associated marginal significance level (p-value), respectively. The numbers in the last column are the number of weekly real Wednesday observations used in this paper.

	Acronyms	Mean	Median	Max.	Min.	Std. Dev.	Skewness	Kurtosis	Jarque-Bera Test	J-B test P-value	No of Obs.
Indonesia	IDN	0.001	0.0007	0.8017	-0.363	0.0584	3.9844	62.207	93537.8	< 0.001	629
S. Korea	KOR	-0.0001	-0.0024	0.2047	-0.196	0.0445	0.2981	5.684	198.1	< 0.001	629
Malaysia	MYS	0.0009	0.0015	0.282	-0.2039	0.04	0.1741	9.233	1031.2	< 0.001	625
Philippines	PHL	0.0004	0.0011	0.1334	-0.1519	0.0377	0.0013	4.102	32.2	< 0.001	636
Taiwan	TAI	0.0019	0.0019	0.231	-0.2155	0.05	-0.3	5.54	177.5	< 0.001	625
Thailand	THA	-0.0003	-0.0009	0.2145	-0.1824	0.0481	0.0938	4.727	79.9	< 0.001	636
Argentina	ARG	0.0039	0.0025	0.4855	-0.3404	0.0798	0.7132	8.898	975.7	< 0.001	636
Brazil	BRA	0.0024	0.0073	0.2985	-0.4151	0.0705	-0.722	7.464	585.2	< 0.001	636
Chile	CHL	0.002	0.0018	0.1269	-0.1488	0.0335	-0.0052	4.889	94.9	< 0.001	638
Mexico	MEX	0.0026	0.0011	0.2053	-0.1853	0.0419	0.0977	5.219	131.9	< 0.001	638
Greece	GRC	0.0025	0.0005	0.1754	-0.1764	0.0445	0.0602	4.346	48.6	< 0.001	638
Turkey	TUR	0.0011	-0.0047	0.2727	-0.2584	0.0739	0.2016	4.12	37.0	< 0.001	626
USA		0.0023	0.0036	0.0662	-0.0915	0.0194	-0.6311	4.951	133.2	< 0.001	592
Japan		0.0004	0.0003	0.0579	-0.0532	0.0145	0.1979	4.743	78	< 0.001	586

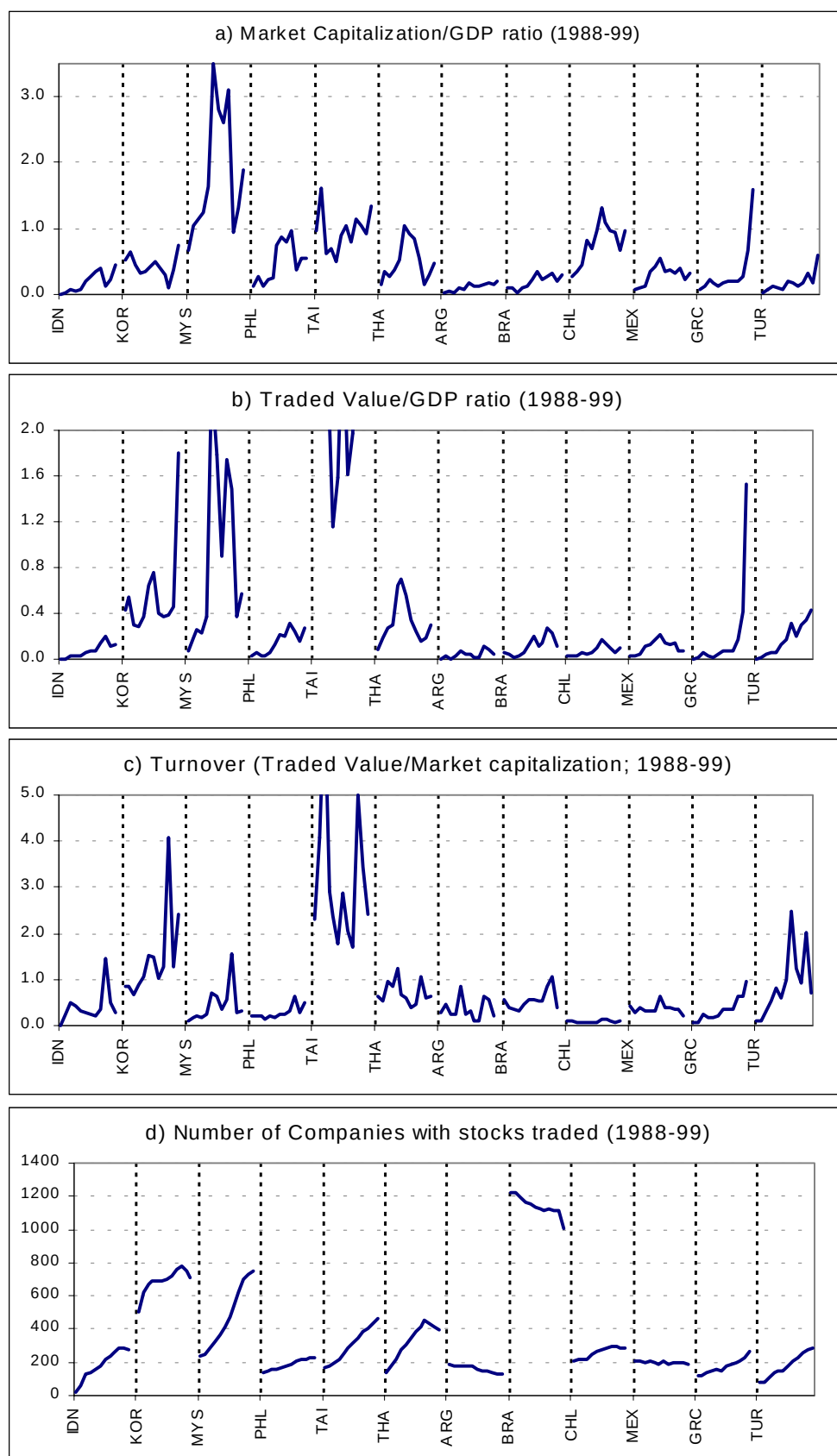
Table 4. **Variance-ratio-based Multiple Comparison Test Statistic** (weekly real Wednesday stock returns)

	Variance Ratio					VR test statistic (z_2)					MCT
	2 weeks	4	8	16	32	2 weeks	4	8	16	32	Statistic
Indonesia	1.062	1.262	1.483	1.504	1.72	0.395	1.07	1.614	1.476	1.858	1.858
S. Korea	0.909	1.034	1.116	1.212	1.388	-1.547	0.324	0.722	0.913	1.195	1.547
Malaysia	0.988	1.007	1.184	1.465	1.637	-0.195	0.058	1.004	1.755	1.758	1.758
Philippines	1.086	1.269	1.416	1.585	1.708	1.737	2.954	3.02	2.948	2.523	3.02 [#]
Taiwan	1.066	1.233	1.347	1.513	1.578	1.245	2.296	2.156	2.13	1.682	2.296 [#]
Thailand	1.106	1.311	1.513	1.622	1.404	2.159	3.277	3.469	2.871	1.31	3.469 [*]
Argentina	1.037	1.098	1.069	1.04	1.018	0.568	0.788	0.36	0.146	0.046	0.788
Brazil	1.109	1.185	1.06	0.911	0.817	1.471	1.503	0.349	-0.386	-0.597	1.503
Chile	1.099	1.277	1.44	1.447	1.377	2.085	3.177	3.3	2.293	1.368	3.3 [*]
Mexico	1.019	1.066	1.14	1.151	1.104	0.324	0.616	0.891	0.666	0.343	0.891
Greece	1.055	1.17	1.377	1.541	1.445	1.114	1.859	2.649	2.593	1.493	2.649 [*]
Turkey	1.115	1.231	1.359	1.397	1.405	2.579	2.735	2.695	2.039	1.469	2.735 [*]
USA	0.93	0.91	0.79	0.75	0.58	-1.58	-1.05	-1.51	-1.24	-1.45	1.58
Japan	0.98	1.11	1.12	1.09	1.09	-0.33	1.15	0.75	0.40	0.28	1.15

#: Random walk hypothesis is rejected at 10% significance level.

*: Random walk hypothesis is rejected at 5% significance level.

Figure 1. **Stock market development indicators for 12 emerging markets ***



* For country name acronyms see Table 3.

Source: IFC Emerging Markets Factbook

Figure 2. **Multiple Comparison test for daily nominal returns in New York and Tokyo Stock Exchanges (January 1988- May 1999)**

The first grey dot on top of each country's name is the MCT statistics for the full January 1988- May 1999 sample (reported in Table 3). Every following grey dot corresponds to a subsample with five daily observations being dropped from the beginning of the sample at a time until a minimum subsample size of 750 daily observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 750-day long subsample with the starting date set in January 1988. Every following solid dot corresponds to a subsample with five daily observations being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

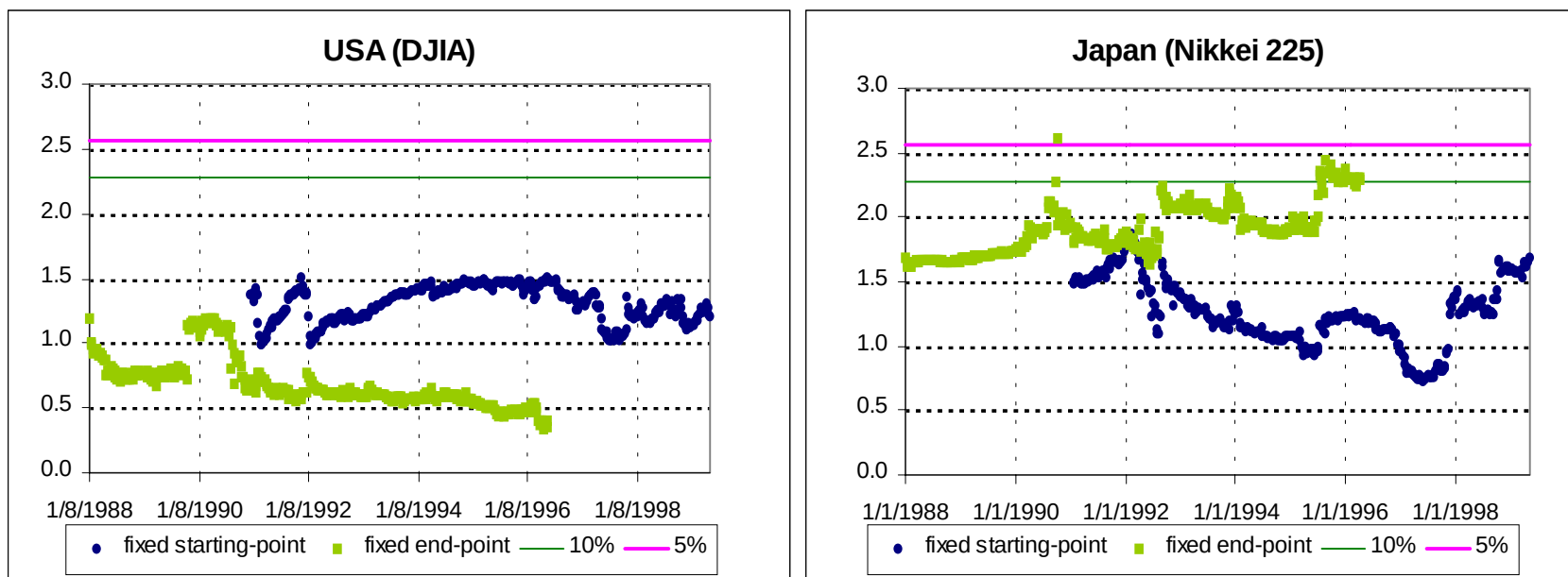


Figure 3. **Multiple comparison test for NYSE & AMEX stocks- Weekly (Wednesday-to-Wednesday) and daily returns for value-weighted index excluding dividends (July 1962 - May 1999)**

The first grey dot on top of each country's name is the MCT statistics for the full July 1962 – May 1999 sample. Every following grey dot corresponds to a subsample with one weekly observation (five daily observations) being dropped from the beginning of the sample at a time until a minimum subsample size of 250 (750) observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 250-week (750-day) long subsample with the starting date set in July 1962. Every following solid dot corresponds to a subsample with one weekly observation (five daily observations) being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

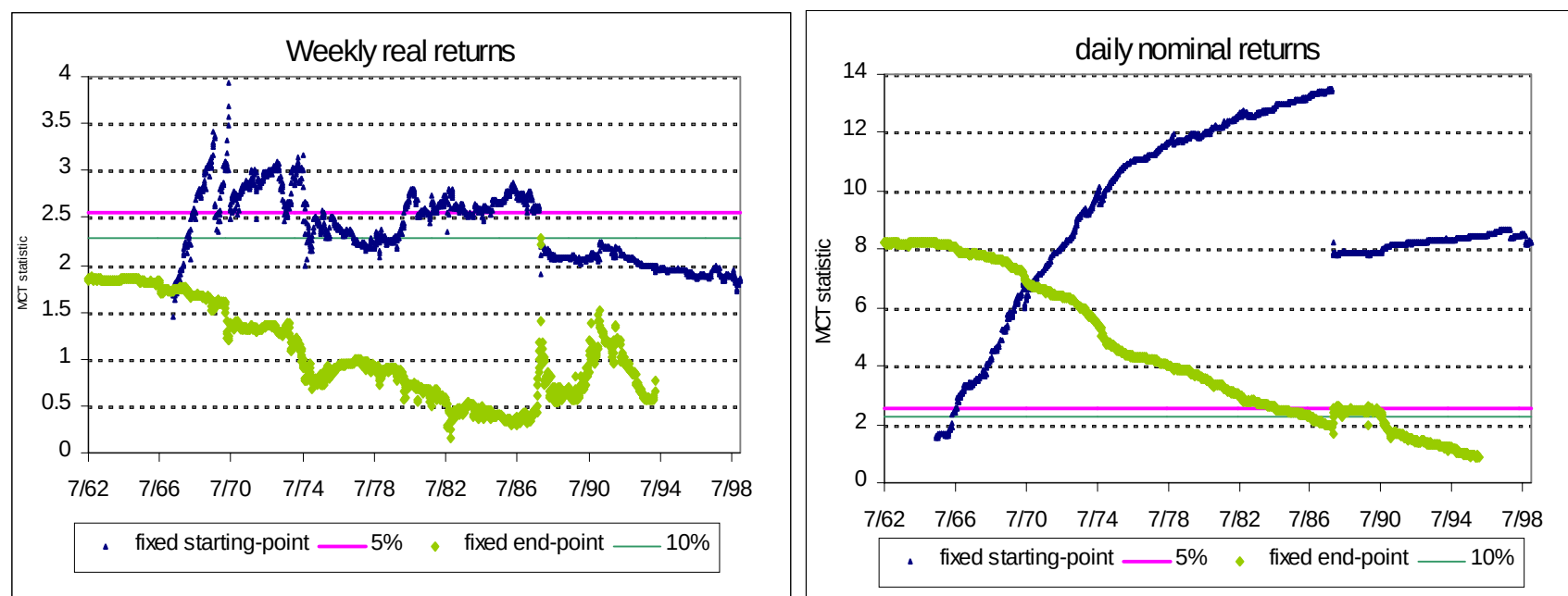


Figure 4. Multiple Comparison test for weekly real Wednesday-to-Wednesday returns in 12 Emerging Markets (Jan. 1988- Mar. 2000)

The first grey dot on top of each country's name is the MCT statistics for the full 1988-2000 sample (reported in Table 3). Every following grey dot corresponds to a subsample with one weekly observation being dropped from the beginning of the sample at a time until a minimum subsample size of 250 observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 250-week long subsample with the starting date set on January 6, 1988. Every following solid dot corresponds to a subsample with one weekly observation being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

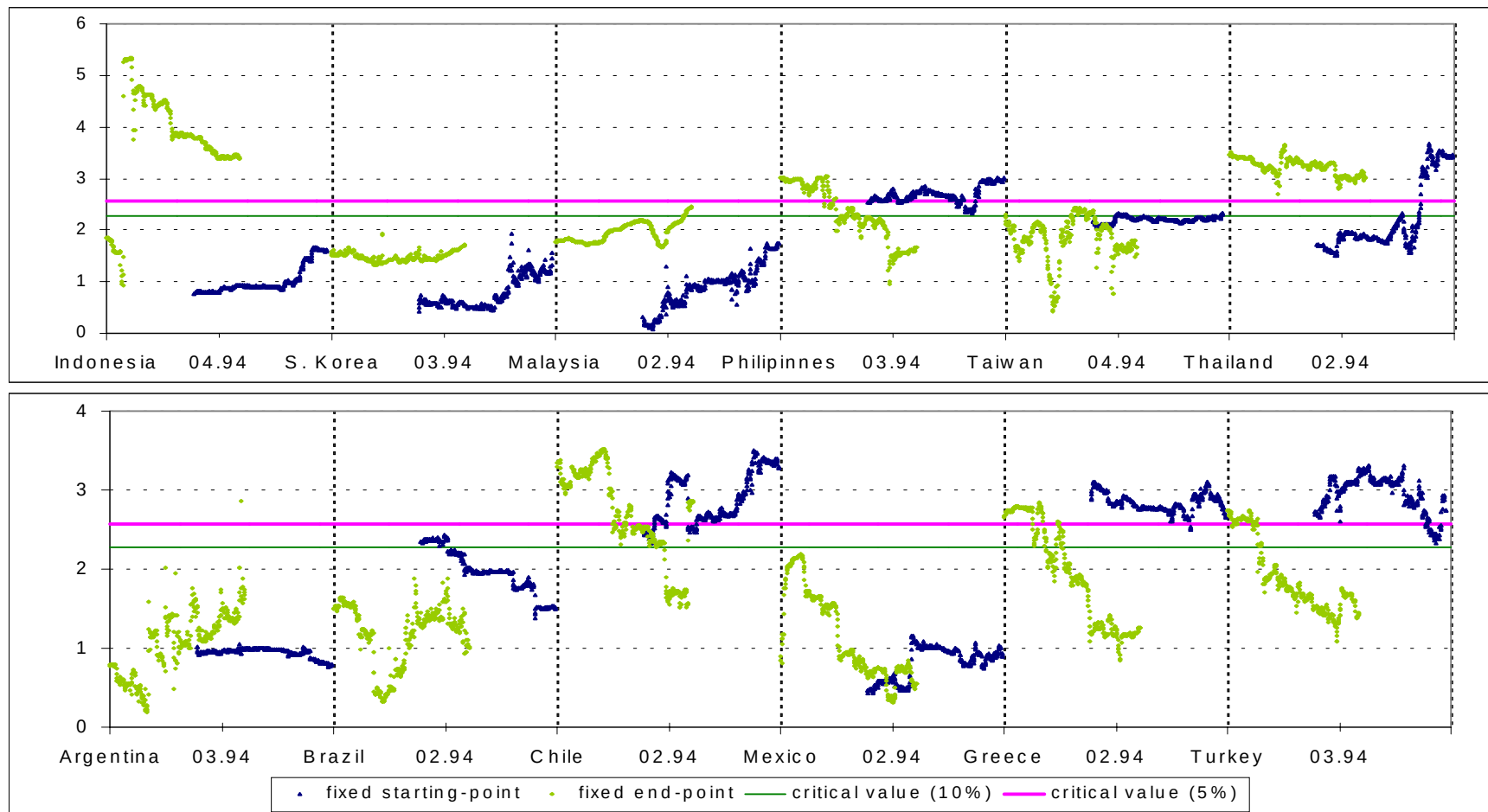


Figure 5. Multiple Comparison test for weekly real Friday-to-Friday returns in 12 Emerging Markets (January 1988- March 2000)

The first grey dot on top of each country's name is the MCT statistics for the full 1988-2000 sample. Every following grey dot corresponds to a subsample with one weekly observation being dropped from the beginning of the sample at a time until a minimum subsample size of 250 observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 250-week long subsample with the starting date set on January 8, 1988. Every following solid dot corresponds to a subsample with one weekly observation being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

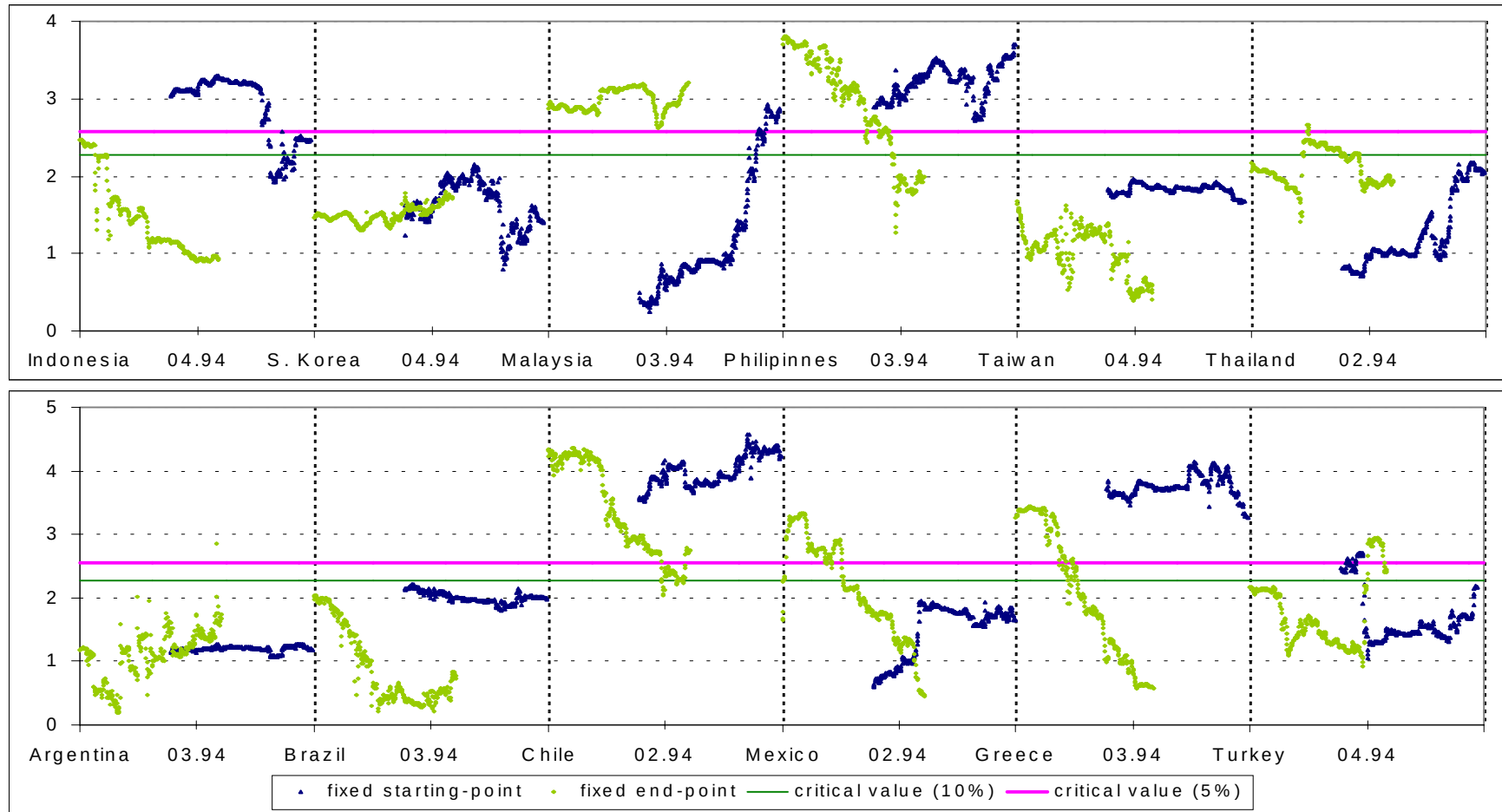


Figure 6. **MCT statistics for weekly real Wednesday-to-Wednesday returns in 12 Emerging Markets (Jan. 1988- Mar. 2000) after dropping the outliers**

The first grey dot on top of each country's name is the MCT statistics for the full 1988-2000 sample. Every following grey dot corresponds to a subsample with one weekly observation being dropped from the beginning of the sample at a time until a minimum subsample size of 250 observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 250-week long subsample with the starting date set on January 8, 1988. Every following solid dot corresponds to a subsample with one weekly observation being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

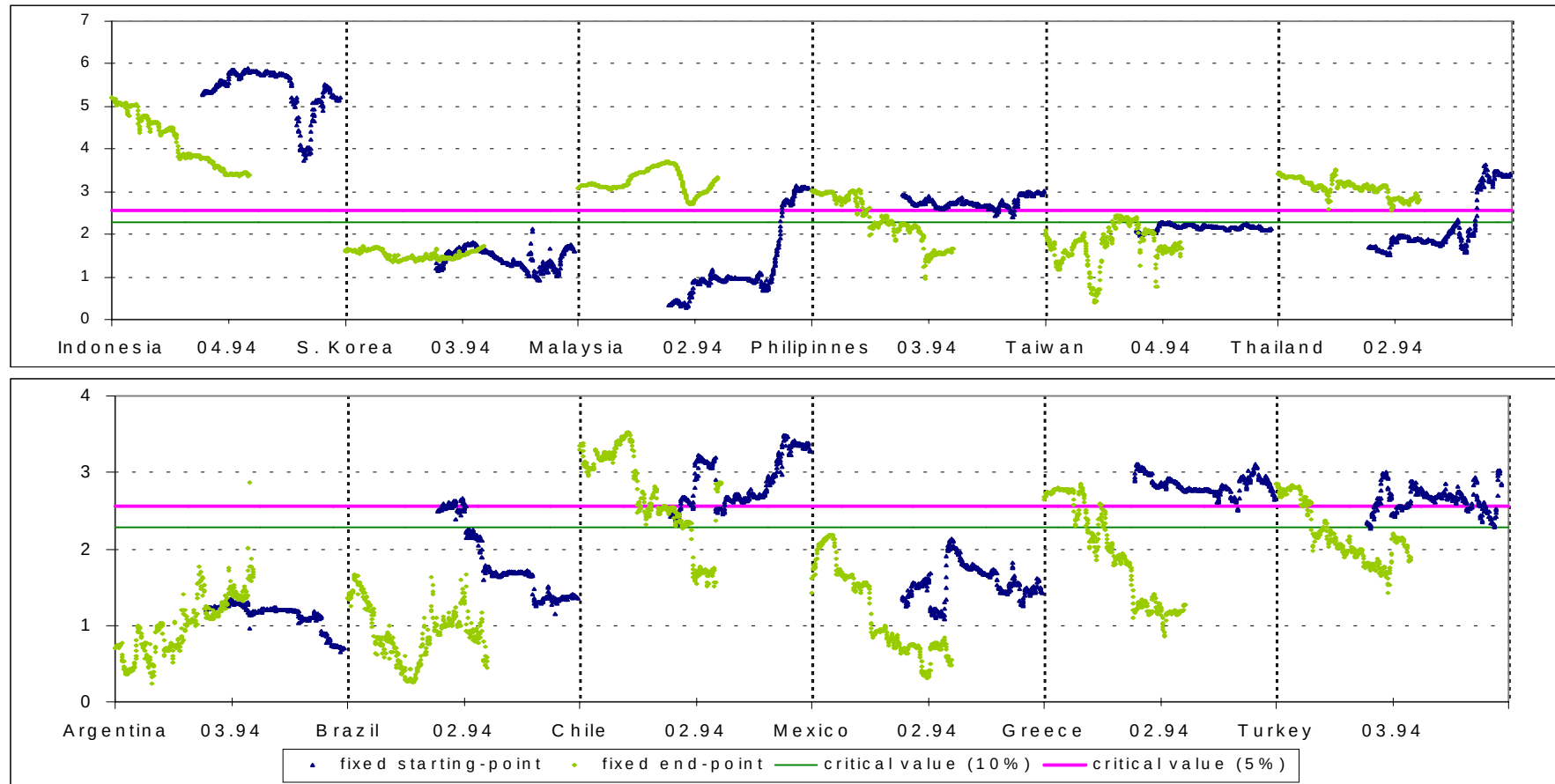


Figure 7. **MCT statistics for weekly real Friday-to-Friday returns in 12 Emerging Markets (Jan. 1988- Mar. 2000) after dropping the outliers**

The first grey dot on top of each country's name is the MCT statistics for the full 1988-2000 sample. Every following grey dot corresponds to a subsample with one weekly observation being dropped from the beginning of the sample at a time until a minimum subsample size of 250 observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 250-week long subsample with the starting date set on January 8, 1988. Every following solid dot corresponds to a subsample with one weekly observation being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

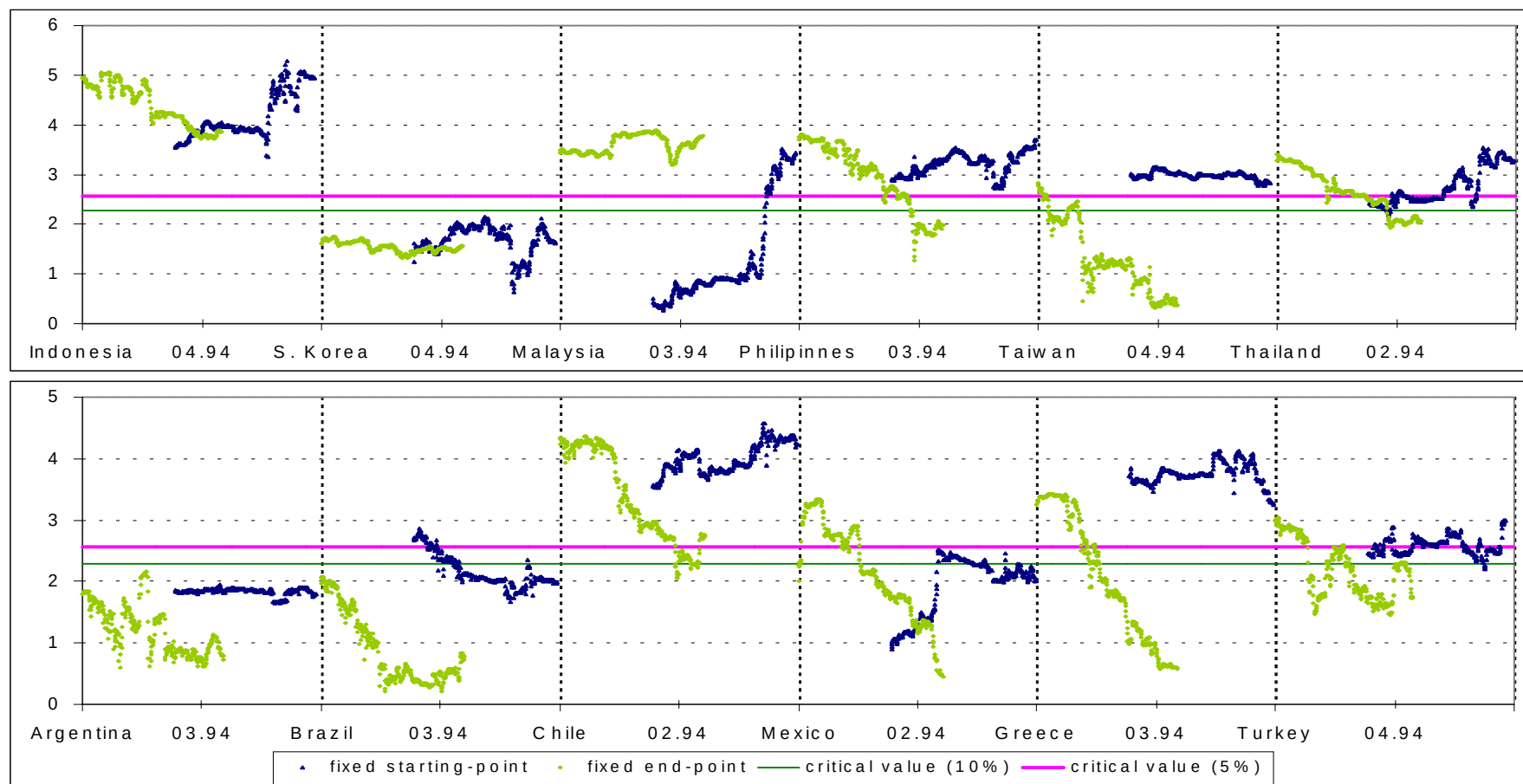


Figure 8. **MCT Statistics for daily nominal returns in 12 Emerging Markets (January 1988- March 2000)**

The first grey dot on top of each country's name is the MCT statistics for the full 1988-2000 sample. Every following grey dot corresponds to a subsample with five daily observations being dropped from the beginning of the sample at a time until a minimum subsample size of 750 observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 750-week long subsample with the starting date set on January 4, 1988. Every following solid dot corresponds to a subsample with five daily observations being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.

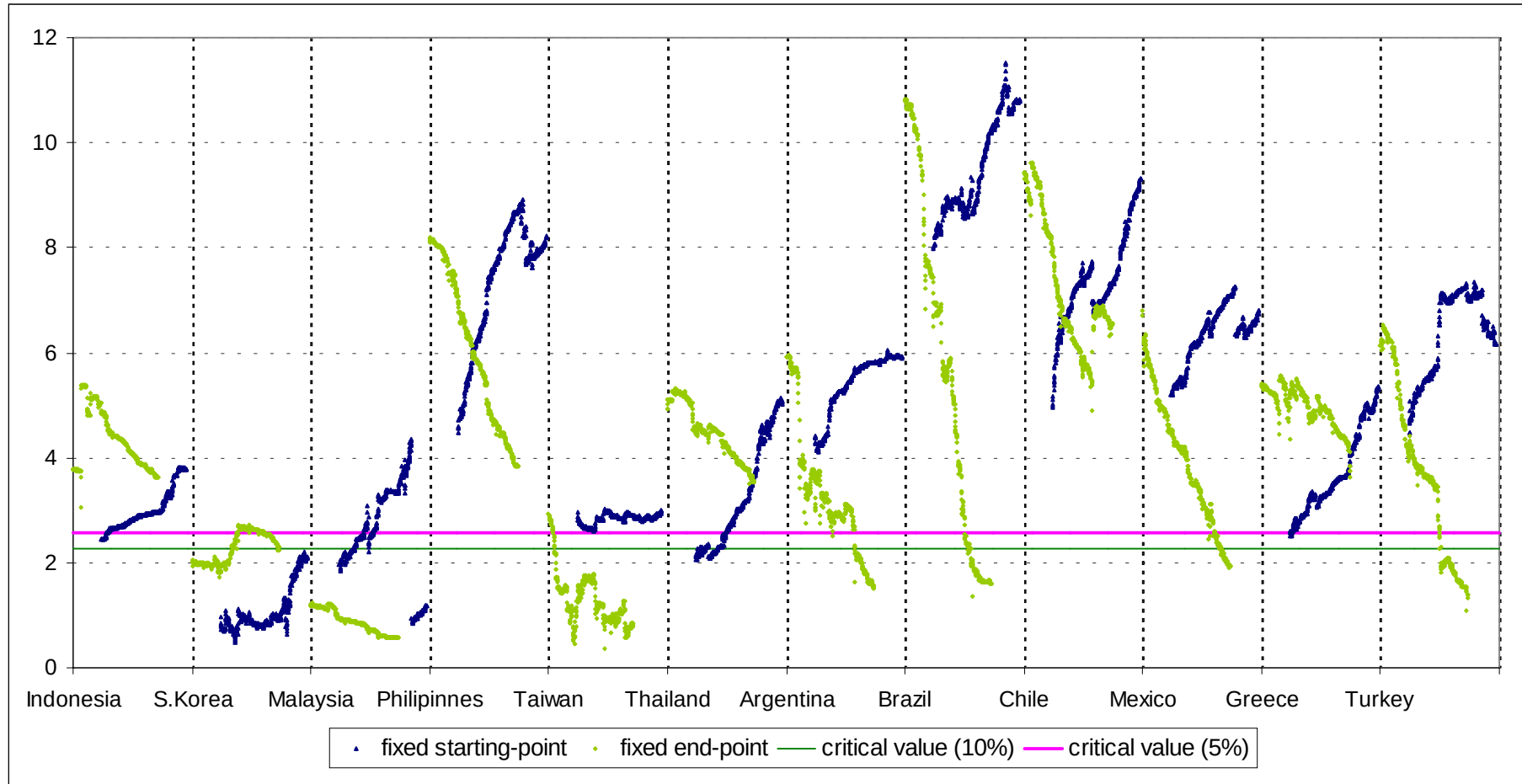
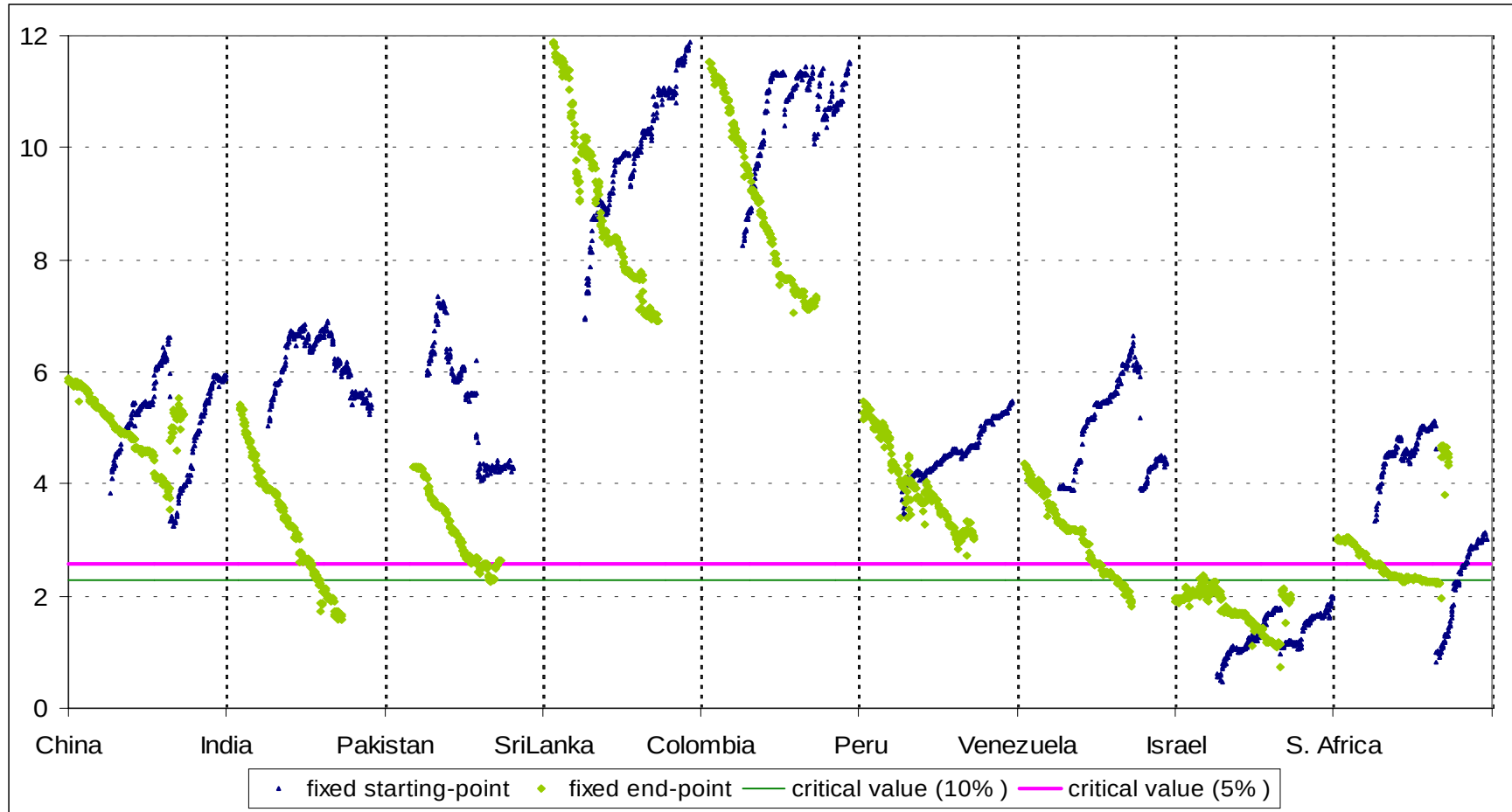


Figure 9. **Multiple Comparison test for daily nominal returns in 9 Emerging Markets (January 1993- March 2000)**

The first grey dot on top of each country's name is the MCT statistics for the full 1988-2000 sample. Every following grey dot corresponds to a subsample with five daily observations being dropped from the beginning of the sample at a time until a minimum subsample size of 750 observations is reached. The first solid dot, on the other hand, is the MCT statistics for the 750-week long subsample with the starting date set on January 4, 1993. Every following solid dot corresponds to a subsample with five daily observations being added to the sample at a time, until the full sample is reached. The bold and normal straight lines indicate the critical values for the MCT statistic at the 5 and 10 percent levels, respectively.



Data Appendix

January 1, 1988 – March 24, 2000:

MSCI Local Currency Index (number of daily observations in parenthesis):

Indonesia (2738), S. Korea (2790), Malaysia (2800), Philippines (2822), Taiwan (2746), Thailand (2779), Argentina (2814), Brazil (2777), Chile (2829), Mexico (2807), Greece (2834), Turkey (2812).

January 1, 1988 – May 5, 1999:

U.S.: Dow Jones Industrial Average Index (2867)

Japan: Nikkei 225 Index (2795)

January 1, 1993 – March 24, 2000:

MSCI Local Currency Index: Israel (1864), South Africa (1806), China (1854), India (1700), Pakistan (1519), Sri Lanka (1729), Colombia (1755), Peru (1805), Venezuela (1764).